

UCS537: SOURCE CODE MANAGEMENT

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2	0	2	3.0

Course Objectives: The objective of the course is to teach techniques to combine software development and IT operations using DevOps. It helps to understand faster software development practices with higher quality.

Traditional Software Development: The Advent of Software Engineering, Waterfall method, Developers vs IT Operations conflict.

Rise of Agile methodologies: Agile Vs Waterfall Method, Iterative Agile Software Development, Individual and team interactions over processes and tools, working software over comprehensive documentation, Customer collaboration over contract negotiation, responding to change over following a plan

Definition and Purpose of DevOps: Introduction to DevOps, DevOps and Agile, Minimum Viable Product, Application Deployment, Continuous Integration, Continuous Delivery **CAMS** (Culture, Automation, Measurement and Sharing): CAMS – Culture, Automation, Measurement, Sharing, Test-Driven Development, Configuration Management, Infrastructure Automation, Root Cause Analysis, Blamelessness, Organizational Learning.

Typical Toolkit for DevOps: Introduction to continuous integration and deployment, Version control system

Source Code Management History and Overview: Examples - SVN, Mercury and Git, History - Linux and Git by Linus Torvalds,

Version Control System: Version control system vs Distributed version control system: Local repository, Advantages of distributed version control system, The Multiple Repositories Models, completely resetting local environment, Revert - cancelling out changes.

Laboratory work:

Basic structure and Implementation of various distributed version control systems for source code management.

Course Learning Outcomes (CLOs) / Course Objectives (COs):

After completion of this course, the students will be able to:

1. Identify the need for migrating from traditional software development to Agile model and then to DevOps.
2. Define and understand the basic principles and need of DevOps and Continuous Delivery.
3. Understand the history and overview of Source Code Management, along with real-time examples.
4. Differentiate between centralized and distributed version control systems and basic operations in version control systems and demonstrate the use of various version control systems.

Text Books:

1. The DevOps Handbook - Book by Gene Kim, Jez Humble, Patrick Debois, and Willis Willis.
2. Pro Git – Book by Scott Chacon and Ben Straub (available at <https://git-scm.com/book/>).

Reference Books:

1. What is DevOps? - by Mike Loukides.

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Evaluation Scheme:

S.No.	Evaluation Elements	Weightage (%)
1.	MST+EST	70
2.	Sessional (May include Assignments//Quizzes)	30

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